

Express Riddler

25 March 2022

Riddle:

You have a case with 11 golden globes, weighing 1 kilogram, 2 kg and so on, up to 11 kg. And you know exactly which globe is which.

You have arranged to sell one of the globes to a queen. She has heard tales of these orbs, and knows for a fact that they have masses from 1 kg to 11 kg. However, she doesn't know which is which and will not simply take your word for it. She will purchase a globe if you can demonstrate its weight.

The queen has a balancing scale with two plates, one on each side. It shows whether the total weight on either plate is equal or, if not, which side is heavier. The queen can clearly see which globes you place on which plate. However, if at any point the mass on either side exceeds 11 kg, the scale will break and the queen will refuse to buy from you.

The queen is in a hurry for her next appointment, so time is limited. What is the fewest number of weighings by which you can prove the weight of at least one globe? Which globe is it?

Solution:

It turns out you can use a single weighing to prove the weight of one globe. By placing the four lightest globes (1, 2, 3, and 4 kg) on one side, their total weight is 10 kg. This is the smallest possible weight of any four globes. If the 11 kg globe is placed on the other side, it will outweigh the other four, thus proving its weight is strictly larger than 10 kg. So the single globe which outweighs four others must be the **11-kg globe**.